

Real-Time Corneal & Retinal Screening

An Edge-AI, Blockchain, and ABDM/ABHA-Integrated Ophthalmic Screening Architecture

Major Project Topic Evaluation & Technical Report

Prepared: July 30, 2026

What this report covers: a plain-language breakdown of the proposed system, the full technical architecture and workflow, a detailed tech stack, target users, a literature comparison table against real published research, an honest feasibility assessment for publishing a research paper from this topic, and concrete novelty angles you can pursue.

1. Executive Summary

Your professor has handed you a genuinely good, timely, and ambitious topic. In plain terms, it asks you to build a system that (a) looks at a photo of someone's eye and tells them, right there on the device, whether they might have diabetic retinopathy, a corneal ulcer, or glaucoma; (b) proves that this AI verdict wasn't tampered with, using blockchain; and (c) safely hands that result into India's national digital health system (ABDM/ABHA) so it becomes part of the patient's permanent, portable medical record — with the patient's explicit consent controlling who sees it.

Each of the three pillars — edge AI ophthalmology, blockchain-secured health records, and ABDM/FHIR interoperability — is individually well-studied in research literature (see Section 12). What is *not* common is combining all three into one coherent, India-specific, real-time pipeline that goes from "photo taken" to "specialist notified" in one automated flow. That combination, plus a couple of deliberate technical choices explained in Section 13, is where your novelty and your publishable contribution can come from.

This is a large, multi-layered project. It is very achievable as a final-year major project if you scope it correctly: build the AI models and the blockchain logging layer for real, and *simulate* the ABDM Sandbox and hospital federated learning parts (which is standard practice, since production ABDM access requires NHA empanelment that a student project will not get). Section 14 gives you a realistic week-by-week plan.

2. Key Technical Terms, Explained Simply

Before diving into the architecture, here's a quick plain-language reference for the core AI/ML and blockchain terms used throughout this report. If you already know some of these, skim past them — this is here so nothing in the later sections assumes knowledge you don't have yet.

2.1 AI, ML, DL, CNN — how these terms relate to each other

These get thrown around like they're all the same thing, but they're actually nested inside each other, each one a more specific version of the last:

- **AI (Artificial Intelligence)** — the broadest umbrella term for any computer program that does something "smart." Too general to describe what you're actually building.
- **ML (Machine Learning)** — a *type* of AI where, instead of a programmer writing exact fixed rules, the computer is shown lots of examples and figures out the pattern itself.
- **DL (Deep Learning)** — a *type* of ML that uses "neural networks" (many stacked layers of simple math units, loosely modeled on how brain cells connect). This is the technique that's actually good at messy real-world inputs like images.
- **CNN (Convolutional Neural Network)** — a *specific kind* of deep learning network built specifically for images. This is what your project actually uses for all three disease checks.
- **NLP (Natural Language Processing)** — a completely separate branch of ML/DL for *text and language* (chatbots, translation, etc.). Your project has nothing to do with NLP — you're working with eye photos, not sentences. Mentioned here only so the term doesn't cause confusion later.

So, in one line: your project = AI → ML → Deep Learning → CNN. That's the whole chain you need to keep in mind.

2.2 What "training a model" actually means

A **dataset** here just means a folder of eye photos where each photo already has the correct answer attached (e.g., "this one is Mild DR," labelled by a real doctor beforehand). **Training** is the process of showing the CNN thousands of these photo+answer pairs; the network starts by guessing randomly, checks how wrong it was, and slightly adjusts its internal numbers to be less wrong next time — repeated thousands of times until it gets good at spotting the real pattern. The end result — the tuned, ready-to-use network — is called a **model**. When this report says "train a model on Dataset X," this is exactly what's meant.

Two model "shapes" show up in this project, doing two different jobs:

- **Classification** — the model picks one label out of a fixed list (e.g., picking one of the 5 DR stages). Used for DR and glaucoma.
- **Segmentation** — the model doesn't just pick a label, it colors in *exactly which pixels* of the image belong to the problem area, like using a highlighter on the image. Used for corneal ulcers (via a CNN variant called **U-Net**, purpose-built for this pixel-by-pixel task).

2.3 Edge AI vs Cloud AI

This isn't a different kind of AI model — it's about *where the same trained model runs*:

- **Cloud AI**: photo is taken → sent over the internet to a remote server → server runs the model → sends the answer back. Needs internet; has some delay.
- **Edge AI**: the same trained model, compressed down and installed directly on the phone/device itself. No internet needed — the phone's own processor does the work, instantly, on the spot.

The tools that do this compressing are **TFLite (TensorFlow Lite)** and **ONNX Runtime** — think of them as converting a large, detailed reference model into a smaller, slightly-less-detailed "pocket version" that still fits and runs on a phone. **Quantization**, mentioned a few times in this report, is one specific compression technique — it stores the model's internal numbers using fewer digits of precision, making the model smaller and faster at a small, usually acceptable, accuracy cost.

2.4 Blockchain, in the simplest possible terms

A **blockchain** is just a shared record book (a ledger) that, once something is written into it, cannot be secretly edited or deleted afterward — every entry is locked in and any tampering becomes obvious. It has nothing to do with cryptocurrency by default; that's just the most famous use of the same idea. Two more terms that come up constantly:

- **Hash** — think of it as a unique fingerprint generated from a piece of data. If even one character of the original data changes, the fingerprint changes completely. This is how the blockchain "seals" a result — it doesn't store the whole medical result on-chain, just its fingerprint, so any later tampering with the real result is instantly detectable.
- **Smart contract** — a small program that lives on the blockchain and automatically runs a simple "if this happens, then do that" rule, with no human needing to trigger it manually. Example used throughout this report: "if the AI's confidence for a critical condition crosses X%, automatically notify the Consent Manager."

There are **public blockchains** (like the real Ethereum or Bitcoin networks — open to millions of strangers, and every write costs a real transaction fee) and **private/permissioned blockchains** (software you run yourself, on your own laptop or college server, with no strangers involved and no real fees). This report recommends the private kind for your project — completely free, and more realistic to how real hospitals would deploy this anyway.

2.5 Federated Learning, in one sentence

Normally, to make an AI model smarter you'd gather everyone's images in one place and train on all of them together. **Federated learning** avoids that: each hospital trains the model using only its own local images, and only shares the resulting math (not the images themselves) with a central coordinator, which blends everyone's updates into one improved model. No hospital ever sees another hospital's patient photos.

3. Background: Why This Problem Matters

2.1 The clinical problem

Diabetic retinopathy (DR) is damage to the blood vessels of the retina caused by long-term high blood sugar, and it is one of the leading causes of preventable blindness worldwide. It is graded into five stages: No DR, Mild, Moderate, Severe, and Proliferative DR. The disease is often called a "silent disease" because it shows no symptoms until it is advanced — *it is asymptomatic, making early detection critical, and traditional screening relies on costly, bulky fundus cameras that are hard to access in remote areas*. Corneal ulcers are open sores on the cornea, usually from infection or injury, diagnosed by staining the

eye with fluorescein dye under blue light so the damaged area glows green *while the healthy area stays unstained*; left untreated, they can scar the cornea and cause permanent vision loss. Glaucoma, the third disease this system targets, damages the optic nerve (often visible as abnormal "cupping" of the optic disc in a fundus photo) and is a leading cause of irreversible blindness globally.

All three conditions share a common structural problem: diagnosis traditionally requires an ophthalmologist, a slit-lamp or fundus camera, and a clinic visit. In India and other low-resource settings, ophthalmologist density is low and concentrated in cities, so people in rural or peri-urban areas either don't get screened or get screened too late.

2.2 Why "edge AI" instead of cloud AI

Sending a photo to a cloud server for analysis works fine with reliable broadband, but rural primary health centres frequently have poor or no internet connectivity. Running the AI model directly on the capturing device (a smartphone, a slit-lamp attachment, or a camera module) removes that dependency entirely — the verdict is available in real time, offline, at the point of care. This is a well-established and clinically validated approach: *one Mumbai-based study of 1,378 diabetic patients found an offline AI algorithm running on a smartphone gave immediate results even where internet access was limited, which helped health workers counsel patients on the spot.*

2.3 Why blockchain and ABDM/ABHA

India's Ayushman Bharat Digital Mission (ABDM) is the government's push to give every citizen a portable, unified digital health record. *ABDM aims to create digital health records for 1.4 billion Indians by 2030, and more than 45 crore ABHA IDs had already been created as of March 2026.* Every hospital, lab, and health app is increasingly expected to plug into this ecosystem. The blockchain layer in your project isn't there for cryptocurrency — it exists to solve a specific trust problem: once an AI makes a diagnosis, how do you prove later that the result shown to the patient is exactly the result the model produced, and that nobody quietly edited it? A tamper-evident ledger answers that. This concern is not hypothetical in medical AI — *researchers studying blockchain in ophthalmology have argued that blockchain's immutability and traceability can increase the validity and transparency of deep learning algorithms in medicine, and can help defend federated learning systems against malicious tampering.*

4. What Exactly Is Being Proposed (In Simple Terms)

Strip away the jargon and the project is four Lego blocks stacked on top of each other:

1. **A smart camera that thinks for itself.** A phone, a slit-lamp attachment, or a small camera module takes a picture of the eye (either a fundus photo of the retina, or a fluorescein-stained photo of the cornea) and a small, efficient AI model running *on that same device* — not in the cloud — instantly decides: "this looks like Moderate DR" or "there's a 12% ulcer area on the cornea" or "the optic disc looks suspicious for glaucoma."
2. **A tamper-proof receipt for that verdict.** The moment the AI gives its answer, a smart contract (a small self-executing program on a blockchain) writes down a fingerprint (a "hash") of that result, a timestamp, and which AI model version produced it. Nobody — not even the hospital's own IT admin — can quietly go back and edit that verdict without it being obvious.
3. **A safe handover into the patient's permanent medical file.** The verdict gets repackaged into a standard medical-record format (FHIR) and pushed into the patient's ABHA Health Locker — India's version of a personal, portable digital health folder that follows the patient between hospitals for life. This only happens with the patient's consent, tracked through ABDM's official Consent Manager.
4. **A way for hospitals to make the AI smarter together, without sharing patient photos.** Multiple hospitals can help train a better version of the AI model using "federated learning" — each hospital trains on its own local images and only shares the resulting math (model weight updates), never the actual patient photos. A smart contract keeps score of who contributed how much training data and can trigger a reward for participating.

If a screening result is dangerous (e.g., signs consistent with retinal detachment, an emergency), the system doesn't just quietly log it — it automatically pings the ABDM Consent Manager to ask the patient's permission to instantly share their full eye-history with an available specialist, so a referral can happen the same day instead of after a week of paperwork.

5. Detailed Technical Architecture

The system is best understood as four layers, each handling one job and handing its output to the next layer. The diagram below shows all four layers and how data moves between them.

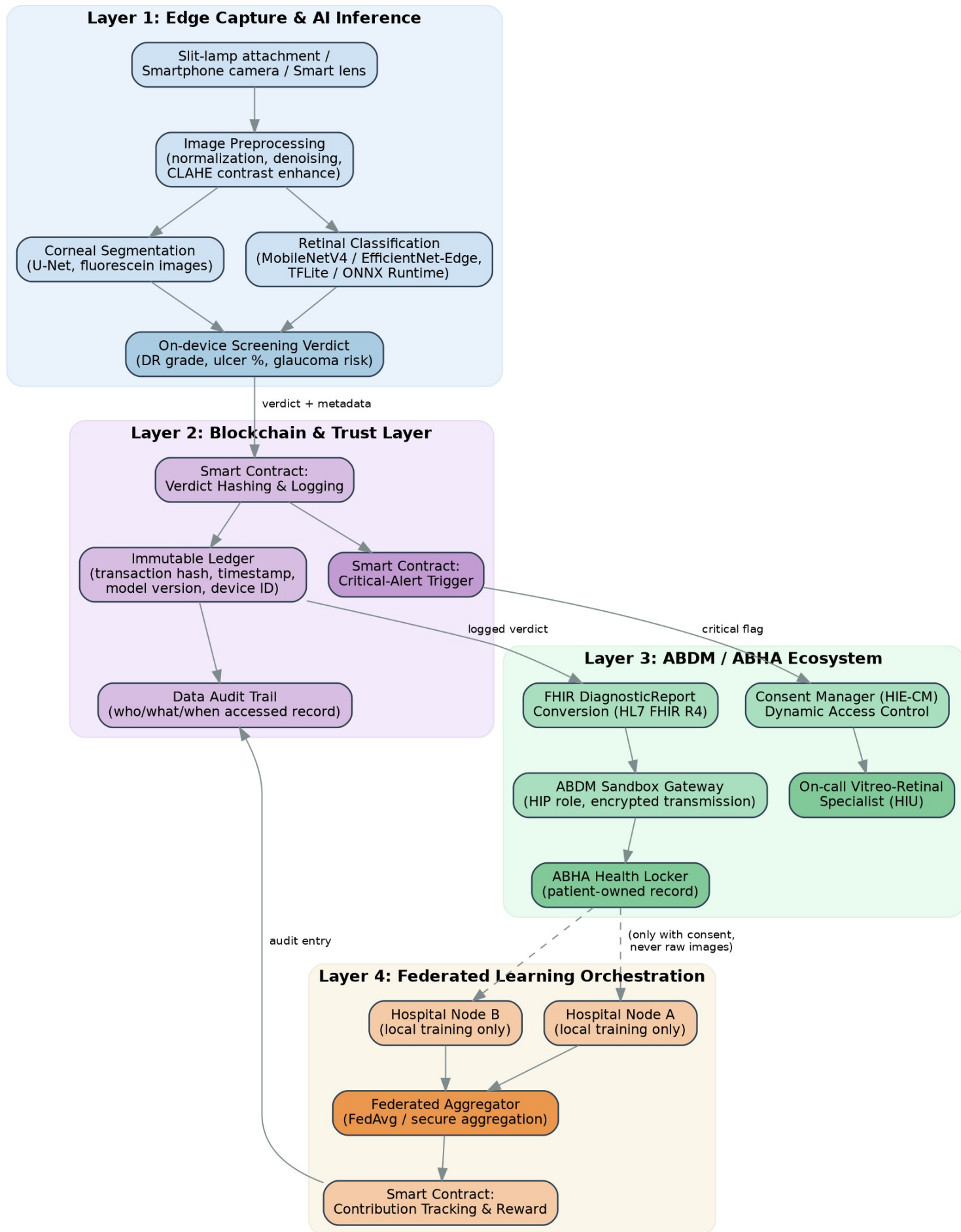


Figure 1. Four-layer system architecture: Edge AI inference → Blockchain trust layer → ABDM/ABHA ecosystem → Federated learning orchestration. Dashed arrows indicate consent-gated, no-raw-data flows.

4.1 Layer 1 — Edge AI Computer Vision Architecture

Goal: produce a clinically useful verdict from a single eye image, on-device, in well under a couple of seconds, without needing an internet connection.

Model selection

The brief specifies lightweight CNNs such as **MobileNetV4** or **EfficientNet-Edge**, converted via **TensorFlow Lite (TFLite)** or **ONNX Runtime** so they can run on constrained hardware. This is a well-validated design choice in the literature: *researchers who deployed and evaluated multiple lightweight CNNs (MobileNet, ShuffleNet, SqueezeNet, and a custom DNN) on edge hardware after quantizing them to 8-bit integers with TensorFlow Lite found MobileNet reached 96.45% accuracy, while SqueezeNet — at only 176 KB — ran in just 17 milliseconds on GPU.* That tells you the accuracy/speed trade-off is real and worth benchmarking yourself rather than assuming.

Practical note: "MobileNetV4" and "EfficientNet-Edge" are reasonable choices to name in your proposal, but for an actual working prototype, MobileNetV3-Small or a quantized EfficientNet-Lite0 are more realistically supported end-to-end by mature TFLite conversion tooling today. Confirm current framework support before committing to a specific version.

Corneal segmentation (U-Net)

For corneal ulcers, the standard clinical technique is fluorescein staining: dye is applied to the eye, the damaged (ulcerated) area absorbs it and glows green under blue/cobalt light, while healthy tissue does not. *Ophthalmologists diagnose ulcer severity by examining images captured this way*, but manual grading is slow and inconsistent between graders. A U-Net — an encoder-decoder CNN originally built for biomedical image segmentation — is well suited to this because it produces a pixel-by-pixel mask of exactly where the ulcer is, not just a yes/no label. Published work on this exact task *reports segmentation performance of Dice similarity coefficient 0.98 for the cornea boundary and 0.97 for the ulcer defect area on the SUSTech-SYSU dataset* (Dice coefficient is just a 0-to-1 score for how well the model's highlighted area overlaps with the doctor's actual marked area — 1.0 would be a perfect match), and a related paper *notes that segmenting corneal ulcers is genuinely hard because of highly variable shapes (point-like vs. flaky ulcers), hazy borders, image noise, and a scarcity of expertly labelled ground-truth images* — which is precisely the public **SUSTech-SYSU dataset** most academic work in this space uses, and which you should also use.

Retinal classification

For DR, the model performs 5-class classification (No DR / Mild / Moderate / Severe / Proliferative) on fundus photographs — the standard public datasets are **EyePACS** (via Kaggle) and **APTOS 2019**, and the **IDRiD** dataset if you also want lesion-level segmentation. For glaucoma, the model instead looks at the optic disc/cup ratio ("cupping") to flag suspicious cases — public datasets include **ORIGA** and **RIM-ONE**. Real deployments back this up clinically: *a prospective clinical validation study in Dominica screening 587 diabetic patients found an AI system achieved 77.5% sensitivity and 91.5% specificity for referable DR compared against a trained field grader, with an AUC of 0.8455* — a useful, honest real-world benchmark to cite rather than the inflated accuracy numbers sometimes seen in lab-only papers.

Engineering tip: build corneal segmentation and retinal classification as two separate model heads/pipelines that share a common preprocessing and TFLite-serving wrapper — don't try to force one multi-task model to do both from day one. Get each working and validated independently first.

4.2 Layer 2 — Blockchain / Smart Contract Layer

This layer does not store medical images on-chain (that would be slow, expensive, and would itself be a privacy violation). Instead, it stores:

- A cryptographic hash of the AI's output (the verdict + confidence score + model version) — this fingerprint changes completely if even one character of the result is altered later.
- A timestamp and a device/session identifier.
- A pointer (not the data itself) to where the full FHIR report lives, e.g., an encrypted reference or an **IPFS** content ID (IPFS is just a free, decentralized file-storage system — used here so the actual report sits off the blockchain, since blockchains are a bad, expensive place to store large files).

A smart contract is simply the code that decides what happens automatically once this data arrives — e.g., "if the confidence score for Proliferative DR or retinal detachment exceeds threshold X, immediately fire a `RequestUrgentConsent` event." This pattern is consistent with the closest matching published work: *a 2023 paper proposes a blockchain-based healthcare framework where a deep learning classifier predicts diabetic retinopathy and the final output is saved into a blockchain architecture that is then accessed by an EHR manager, ensuring only authorized access to the prediction results and related patient information* — this is discussed in detail as your closest prior-art comparison in Section 12.

Realistic choice for a student project: use a permissioned blockchain (Hyperledger Fabric) or a private Ethereum testnet (Ganache/Hardhat + Solidity), not a public mainnet — public chains are unnecessary here (there's no need for anonymous, trustless global consensus among strangers) and cost real transaction fees. A permissioned chain among "hospital nodes" you simulate is both more realistic to what real hospitals would deploy and much cheaper to build and defend in a viva.

4.3 Layer 3 — ABHA/ABDM Integration & Ecosystem Mapping

This is the compliance and interoperability layer, and it has its own vocabulary you should get comfortable with:

Term	What it actually means
ABHA	A 14-digit digital health ID that lets a patient link all their medical records across different providers, labs, pharmacies, and insurers under their own explicit consent.
ABDM	The government infrastructure layer — run by the National Health Authority — that makes those records portable and interoperable across India's whole healthcare system.
HIP	Health Information Provider — systems of record such as labs, hospitals, and pharmacies that create and publish health data. Your screening device/app plays this role.
HIU	Health Information User — consumers of health data, such as insurers or clinics. The on-call specialist's system plays this role when requesting access.
HIE-CM / Consent Manager	The consent management system that mediates every single data exchange — nothing moves without a logged, patient-approved consent artifact.
FHIR R4 (with India IG)	HL7's Fast Healthcare Interoperability Resources standard, which ABDM has adopted with India-specific Implementation Guides that constrain resources to reflect local needs like ABHA identifiers and Indian coding preferences.
Health Locker	The patient's personal digital record store; a HIP publishes structured FHIR R4 records here whenever a clinical event happens.

The practical build sequence, based on how real ABDM integrations are structured, is: *register in the sandbox, choose your role (HIP, HIU, or both), implement ABHA patient-identity flows, integrate the HPR/HFR practitioner and facility registries, build the consent-manager state machine for HIE-CM, exchange records as FHIR R4 bundles using the India-specific profiles, and sign every request with JWTs* — production credentials are a separate, much later step that involves NHA empanelment and is out of scope for a student prototype (more on this in Section 14).

4.4 Layer 4 — Federated Learning Orchestration

Instead of pooling every hospital's retinal images into one central server (a privacy and regulatory nightmare under India's DPDP Act), each hospital trains the shared model locally on its own images, and only the resulting model weight-updates travel to a central aggregator, which merges them (commonly using an algorithm called **FedAvg**, short for "Federated Averaging" — it just means mathematically blending every hospital's updates together into one improved shared model) into an improved global model that's sent back out. Published research validates this is genuinely useful for eye disease models specifically: *a study analyzing multiple federated learning algorithms for glaucoma diagnosis from retinal fundus images found FL approaches can match the performance benefits of centralized training while keeping data at each hospital*, and a large multi-cohort ophthalmology study developed federated deep learning algorithms for detecting myopic macular degeneration from retinal photographs and multi-disease classification from OCT scans, benchmarking against standard algorithms like FedAvg, weighted FedAvg, and FedProx.

The smart-contract "incentive" idea your brief describes — rewarding hospitals for contributing training data/compute without ever seeing raw images — is a real, active research direction combining blockchain's audit trail with FL's privacy guarantee: *researchers argue blockchain's immutability and traceability make it well suited to cryptographically chaining model weight updates in federated learning, both to verify contributions and to guard against malicious tampering with the shared model*.

4.5 Data Auditing Trails

Every time any AI model or any human user (a doctor, a researcher, an insurer via HIU) touches a specific ABHA record, that access is written as an immutable ledger entry: who accessed it, what they accessed, when, and under what consent artifact. This is not a "nice to have" — it's close to a regulatory expectation. *A review of blockchain applications in healthcare notes that patient-centred interoperability, consent tracking, and remote-monitoring audit trails are among the clearest practical use cases already being explored*.

6. End-to-End Workflow (Step by Step)

Walking through a single patient's journey makes the whole system click into place:



Figure 2. Patient journey from registration to specialist referral.

1. **Registration:** the patient's ABHA ID is scanned (QR code) or created on the spot at the screening camp/clinic.
2. **Image capture:** a health worker (who does *not* need to be an ophthalmologist) captures a fundus photo and/or a fluorescein-stained corneal photo using the smart-lens/slit-lamp attachment or a smartphone-based fundus adapter.
3. **On-device inference:** the TFLite/ONNX model runs immediately on the device — no upload, no waiting for signal. Output: DR grade, ulcer segmentation mask + percentage area, glaucoma risk flag, plus a confidence score for each.
4. **Blockchain logging:** the verdict is hashed and written to the smart contract ledger along with model version and timestamp — this becomes the tamper-evident "receipt."
5. **FHIR conversion:** the verdict and metadata are packaged into a FHIR `DiagnosticReport` resource, following ABDM's India-specific implementation guide.
6. **ABDM transmission:** the FHIR bundle is pushed through the ABDM Sandbox gateway (acting as a HIP) and lands in the patient's ABHA Health Locker.
7. **Branch — routine case:** the patient is notified of the result and a routine follow-up interval is suggested (this is where an app UI or SMS notification would sit).
8. **Branch — critical case:** if the verdict crosses a severity threshold (e.g., Proliferative DR, large flaky ulcer, suspected retinal detachment), the smart contract automatically raises a request to the ABDM Consent Manager asking the patient's permission to share their full ophthalmic history with an available on-call vitreo-retinal specialist.
9. **Specialist access:** once consent is granted, the specialist (acting as a HIU) gets time-boxed, logged access to the patient's history and can initiate a teleconsultation or an urgent in-person referral.

Every one of these nine steps produces its own audit-trail entry, so a later compliance review can reconstruct exactly what happened, when, and who authorized it.

7. Complete Technology Stack

Layer	Component	Recommended Technology	Notes
Edge AI / CV	Model training	Python, TensorFlow/Keras or PyTorch	Train on GPU (Colab/Kaggle free tier is enough for a student project)
	Corneal segmentation	U-Net (custom or MONAI implementation)	Use SUSTech-SYSU dataset
	Retinal classification	MobileNetV3/EfficientNet-Lite, transfer learning	Use EyePACS/APTOS/IDRiD; realistic accuracy target 85-92%, not 99%
	On-device runtime	TensorFlow Lite or ONNX Runtime Mobile	Quantize to INT8 post-training for speed
	Capture app	Android (Kotlin) or Flutter cross-platform app; slit-lamp adapter is a 3D-printed/off-the-shelf accessory	Camera2 API for manual exposure control on Android
Blockchain	Platform	Hyperledger Fabric (permissioned) <i>or</i> Ethereum private testnet via Ganache/Hardhat	Fabric is closer to what real hospital consortiums use; Ethereum/Solidity is faster to prototype and demo
	Smart contracts	Solidity (Ethereum) or Chaincode in Go/JS (Fabric)	Keep contracts small: hash-logging + threshold-triggered events
	Off-chain storage pointer	IPFS (for encrypted FHIR bundle references)	Never store raw images or raw FHIR JSON on-chain
	Contract testing	Hardhat/Truffle + Mocha/Chai, or Fabric test network	
ABDM/FHIR	FHIR server	HAPI FHIR (open-source, Java) for local testing	Widely used reference implementation
	ABDM connectivity	ABDM Sandbox APIs (M1: ABHA, M2: Linking, M3: Data exchange)	Free developer sandbox; production access needs NHA empanelment — simulate this for your demo
	Auth	OAuth2 / JWT signing per ABDM gateway spec	
	Backend	Node.js/Express or Python/FastAPI orchestrating FHIR + ABDM + blockchain calls	
Federated Learning	FL framework	Flower (flwr) or TensorFlow Federated	Flower is easier to demo across "simulated hospital" processes/containers
	Aggregation algorithm	FedAvg (baseline), FedProx (for heterogeneous data)	Compare both in your results — this alone is a nice mini-contribution
	Incentive layer	Smart contract tallying rounds/data contributed per node	Can be genuinely simple: an on-chain counter +

Infra / DevOps	Containerization	Docker + docker-compose (one container per simulated hospital node, one for the FHIR server, one for the blockchain node)	threshold-based token mint Makes your demo reproducible in the viva
	Dashboard	React.js admin dashboard showing audit trail, consent status, model performance	Good visual centerpiece for your project demo

8. Target Users & Stakeholders

Stakeholder	What they get from this system
Rural/semi-urban patients	Immediate, offline, low-cost eye screening without needing to travel to a city ophthalmologist first; a permanent digital record of every screening tied to their ABHA ID.
ASHA/community health workers	A simple capture-and-go tool that doesn't require ophthalmology training to operate — the AI does the interpretation.
Primary Health Centres (PHCs)	A screening capability they currently lack, without needing constant internet or an on-site specialist.
Ophthalmologists/vitreo-retinal specialists	Pre-triaged referrals — they only get pinged for genuinely urgent cases, with full consented history instantly available instead of chasing paper records.
Hospitals/research institutions	A way to collaboratively improve the shared AI model via federated learning without ever exposing patient images to each other or to a central party.
NHA/ABDM ecosystem	A concrete, compliant HIP integration example demonstrating consent-driven data flow and audit-ability, aligned with national digital health goals.
Insurers (longer-term, via HIU role)	Consented, verifiable access to diagnostic history for claims — out of scope for your prototype, but worth mentioning as future scope.

9. Datasets Required for Each Component: Links, Access & Limitations

Every model in this project depends on a public dataset, and each one comes with its own access process and real limitations you should state upfront in your report rather than discover mid-project. Here is the exact dataset needed per requirement, where to get it, and what to watch out for.

8.1 Retinal classification (Diabetic Retinopathy grading)

Dataset	Link	Access	Key limitations
APTOS 2019 Blindness Detection	https://www.kaggle.com/c/aptos2019-blindness-detection	Free, instant (Kaggle account + accept competition rules)	<i>Highly imbalanced: roughly 49% No-DR images versus only ~5-8% each for Severe and Proliferative DR</i> , so the model will need class-weighting or oversampling or it will just get lazy and predict "No DR" most of the time.
EyePACS / Kaggle DR Detection	https://www.kaggle.com/c/diabetic-retinopathy-detection	Free, instant (Kaggle account)	<i>Very large (~88,700 images) but images vary wildly in quality/lighting since they come from many different clinics</i> , and a meaningful fraction are low-quality/ungradable — plan for a preprocessing/quality-filtering step.
IDRiD (Indian Diabetic Retinopathy Image Dataset)	https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid	Free, but requires creating an IEEE DataPort account and agreeing to terms	<i>The first dataset representative specifically of an Indian population</i> — very useful for your India-context framing — but small (only 516 images), so use it for fine-tuning/validation rather than training from scratch.

Combined APTOS+IDRiD+Messidor+EyePACS set (pre-merged)	https://www.kaggle.com/datasets/harsha1289/combined-dr-dataset-aptosidridmessidoreyepacs	Free, instant	Convenient single download, but you inherit the class imbalance and quality-variance issues of all four source datasets combined — check the merge methodology before trusting it blindly.
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8.2 Corneal ulcer segmentation

Dataset	Link	Access	Key limitations
SUSTech-SYSU Corneal Ulcer Dataset	https://github.com/CRazorback/The-SUSTech-SYSU-dataset-for-automatically-segmenting-and-classifying-corneal-ulcers (also mirrored at https://doi.org/10.6084/m9.figshare.c.4526675)	Free, direct download (zip file), no registration needed	This is essentially the <i>only</i> public dataset for this exact task, and it's small: <i>712 total images, with only 354 carrying pixel-level segmentation ground truth</i> . This is the single biggest data limitation in your whole project — expect to lean heavily on data augmentation and transfer learning rather than training a large U-Net from scratch.

Important: because the corneal ulcer dataset is so small, do not expect (or promise in your proposal) the same accuracy numbers you'll get on the much larger retinal datasets. Report this limitation honestly — it is a known, published constraint *explicitly called out by other researchers working with this exact dataset, citing hazy ulcer borders, noise, and scarcity of expertly labelled ground truth as the core difficulty*, not something you did wrong.

8.3 Glaucoma detection (optic disc/cup)

Dataset	Link	Access	Key limitations
ORIGA / ORIGA-light	https://www.kaggle.com/datasets/sshikamaru/glaucoma-detection	Free, requires Kaggle registration	<i>Only 650 images total (168 glaucoma-positive, 482 healthy)</i> — small and class-imbalanced toward healthy eyes.
RIM-ONE (r1/r2/r3) and RIM-ONE DL	http://medimrg.webs.ull.es/research/retinal-imaging/rim-one/ ; DL version repo: https://github.com/miag-ull/rim-one-dl	Free, open, for research/educational use only <i>without needing to request permission from the authors, though commercial redistribution is prohibited</i>	<i>Very small (169 images in the classic version)</i> ; images gathered from only three Spanish hospitals, so a model trained purely on this will likely not generalize well to Indian patient demographics or different camera hardware.
DRISHTI-GS	Search "DRISHTI-GS glaucoma dataset" (hosted via IIIT Hyderabad's project page)	Free, registration/request form typically required	<i>Only 101 images</i> , but notably annotated by 5 clinicians with majority-vote grading, useful as a small, high-quality validation set.

Practical recommendation: combine ORIGA + RIM-ONE + DRISHTI-GS for glaucoma (a common strategy in published work) to get a slightly larger, more diverse pool — you'll still be working with a few hundred to ~1,000 images total, so treat glaucoma detection as your smallest-data, highest-transfer-learning-dependency module.

8.4 Cross-cutting dataset issues that apply to all three models

- **Demographic/hardware mismatch.** Almost every public ophthalmology dataset was collected with professional fundus cameras or slit-lamps in specific countries/hospitals — not with a smartphone-based adapter, which is what your Layer 1 device is supposed to use in the field. Expect a real accuracy drop when you test your trained model on photos from your own actual capture device, and budget time to fine-tune on a small set of your own captured images (even simulated/self-photographed test images) rather than assuming lab accuracy will transfer.
- **Label noise and inter-grader disagreement.** DR and glaucoma grading has known human inter-observer variability — even the "ground truth" labels in these datasets were produced by different graders using different protocols across datasets, so don't be surprised by a noise ceiling below 100% no matter how good your model is.
- **No public dataset combines all three conditions in one patient cohort.** You will be training and validating three separate, independent pipelines on three separate datasets — there is no existing multi-disease eye dataset that lets you validate a single combined "one photo, three diagnoses" claim end-to-end. State this explicitly as a scope boundary in your report.
- **Licensing.** Most of these datasets are free for academic/research use but explicitly prohibit commercial redistribution (see RIM-ONE's terms above) — fine for a student project and any resulting paper, but flag this if anyone on your team later imagines turning this into a commercial product.

10. Anticipated Problems & Practical Solutions

Beyond the dataset issues above, here are the specific engineering problems most teams building something like this actually hit, and realistic ways around each one.

Problem	Why it happens	Practical solution
Model works great in Colab/Jupyter but is slow or crashes after TFLite/ONNX conversion on a real phone	Some layers/operations (custom activation functions, certain attention blocks) aren't supported by the mobile runtime, or quantization degrades accuracy more than expected	Pick architectures with known, mature mobile support (MobileNetV3, EfficientNet-Lite) from day one; test conversion on a real low/mid-range Android phone early — in week 2, not week 20; keep a non-quantized fallback for accuracy comparison.
Class imbalance makes the model just predict "No DR" or "Normal" for everything and still score high accuracy	Real disease datasets are naturally imbalanced (most people don't have severe disease)	Use weighted loss functions, oversampling/undersampling, and — critically — report sensitivity/specificity/F1 per class, not just overall accuracy, in your results.
U-Net overfits badly on the small corneal ulcer dataset (354 labelled images)	Too few images for a model with millions of parameters	Heavy data augmentation (rotation, flips, elastic deformation, brightness/contrast jitter appropriate to fluorescein imaging), transfer learning from a pretrained encoder, and k-fold cross-validation instead of a single train/test split so your reported numbers are more trustworthy.
Blockchain writes are slow, and every screening now takes several extra seconds	Consensus mechanisms (even permissioned ones) add latency compared to a plain database write	Log the hash <i>asynchronously</i> — show the patient their result immediately from the on-device model, and let the blockchain write happen in the background without blocking the UI; this is exactly how real systems handle this trade-off.
ABDM sandbox APIs return confusing errors or documentation is incomplete/changes without notice	The sandbox is actively evolving government infrastructure, not a stable commercial API	Budget extra time here specifically (Section 14's roadmap already does); join ABDM developer community/Slack/forum if available; keep a mocked/stubbed version of the ABDM gateway so the rest of your pipeline can be demoed even if the live sandbox is down on viva day.
Federated learning simulation on one laptop is unrealistically fast/clean compared to real distributed hospitals	No real network delay, no real data heterogeneity between "hospital nodes"	Deliberately partition your dataset unevenly across simulated nodes (not a clean random split) to simulate real hospital data heterogeneity, and add artificial latency/dropout to at least one experiment to show you understand the real-world gap — reviewers/examiners respect this honesty.
Smart contract has a bug that lets a "critical" verdict silently fail to trigger the consent request	Threshold logic errors, or the event listener architecture missing an emitted event	Write unit tests for every contract function <i>before</i> integration testing (standard practice with Hardhat/Truffle or Fabric's test network) — test the failure path explicitly, not just the happy path.
Team members work in silos and the four layers don't actually connect at demo time	Classic integration risk on any multi-layer project with a small team	Define your JSON/API contract between layers (what exactly does Layer 1 hand to Layer 2, and Layer 2 to Layer 3) in week 1, before anyone starts coding, and do a rough end-to-end "hello world" pipeline connecting all four layers early, even with dummy data — then improve each layer's real content afterward.
Examiners/reviewers ask "why blockchain instead of a normal signed database log?"	This is the single most common pushback question for any blockchain-in-healthcare project	Have your answer ready: blockchain earns its place specifically when multiple mutually distrusting parties (hospital, patient, auditor, insurer) need to agree on a shared, unforgeable history without a single trusted central admin — for a single-hospital deployment a signed database might genuinely suffice, and it's fine to say so honestly while explaining why the multi-hospital federated learning scenario changes that calculus.

11. Cost Breakdown: What's Free vs. What Involves Money

The good news: this entire project can be built and demoed for a final-year submission at essentially zero cost. Here's exactly where money could enter, so there are no surprises.

Component	Cost	Details
Model training compute	Free	Google Colab free tier or Kaggle Notebooks (both give free GPU hours) are enough for MobileNet/EfficientNet/U-Net training at these dataset sizes.
All datasets listed in Section 9	Free	Every dataset above is free for academic/research use (Kaggle account or free IEEE DataPort/institutional registration at most).
Blockchain platform (Hyperledger Fabric)	Free	Fully open-source; runs locally/on your own or college servers via Docker.
Blockchain platform (Ethereum route instead)	Free	Ganache (local blockchain simulator) and Hardhat/Truffle are free and open-source. <i>Only</i> becomes non-free if you deploy to a real public testnet/mainnet requiring real or test ETH gas fees — avoid this entirely for a student project; stay local.
FHIR server (HAPI FHIR)	Free	Open-source, self-hosted.
ABDM Sandbox access	Free	The public developer sandbox for ABHA creation/linking (M1/M2 milestones) is free to register for and use for development/testing.
Federated learning framework (Flower / TensorFlow Federated)	Free	Open-source.
Android/Flutter app development	Free	Android Studio and Flutter SDK are free; you only need a real Android phone (likely already owned by a team member) for on-device testing.
Slit-lamp attachment or fundus adapter hardware	Possible small cost	Basic smartphone fundus adapters (D-Eye style clip-ons or 3D-printed equivalents) can range from a few thousand to a few tens of thousands of rupees if you want a real physical prototype. <i>Workaround:</i> for the software/AI demo, you don't need one at all — use existing dataset images to simulate "captured" images live during your demo.
Cloud hosting (if you want a live, always-on demo instead of a local one)	Optional cost	AWS/GCP/Azure free tiers cover light use, but can incur real charges beyond free-tier limits. Not needed — running everything locally via Docker Compose for your viva/demo is completely acceptable and often preferred by examiners since it's easier to inspect.
Publishing a paper (if you pursue this per Section 13)	Real cost, only if you publish	Many journals (including MDPI titles like <i>Electronics/Diagnostics</i> , referenced in Section 12) charge an Article Processing Charge (APC), often USD 1,000-2,500+, to publish open-access. IEEE conferences typically charge student registration fees (varies widely, often INR 3,000-15,000 for domestic student conferences). This is separate from the project build itself — many colleges have funds/waivers for student-author papers, so check with your guide/department before assuming you'd pay this personally.
Domain name / production ABDM (HIP) empanelment	Not applicable to a student project	Production-grade ABDM empanelment involves compliance costs meant for real healthcare providers, not student prototypes — you will not need this; the sandbox is the correct and free environment for your entire project.

Bottom line: you can build, run, and demo 100% of this project — all four layers, end-to-end — without spending a single rupee, as long as you (a) train models on Colab/Kaggle free GPUs, (b) keep your blockchain local/permissioned rather than on a public network, (c) use the free ABDM developer sandbox, and (d) skip buying physical slit-lamp hardware in favor of simulating image capture from existing datasets during your demo. The only place real money can enter is optional: buying physical capture hardware for a tangible prototype, or paying journal/conference fees if you later choose to publish — and even the latter is often covered by departmental funds.

12. Literature Review: Comparable Research Papers

No single published paper combines all four layers (edge AI + corneal segmentation + blockchain + ABDM/FHIR + federated learning) the way your brief describes. But every individual piece has close prior art, and one paper in particular — Uppamma & Bhattacharya, 2023 — is your single closest comparison point and should be read in full before you start building.

Paper (Link)	What it says / does	Similarity to your topic	Key difference from your topic
Diabetic Retinopathy Detection: A Blockchain and African Vulture Optimization Algorithm-Based Deep Learning Framework Uppamma & Bhattacharya, <i>Electronics</i> , 2023 mdpi.com/2079-9292/12/3/742	<i>Uses the IEEE DataPort IDRiD dataset, preprocesses with median filtering and lesion segmentation, tunes hyperparameters with a Taylor African Vulture Optimization algorithm, classifies with SqueezeNet, and stores the final output in a blockchain architecture accessed by an EHR manager for authorized access to prediction results.</i>	Closest existing match. Same core idea: DR deep learning model → blockchain-logged output → EHR access control. Confirms the DR+blockchain combination is a validated, publishable research direction.	No edge/on-device deployment (runs on standard compute, not TFLite/ONNX on mobile); no corneal ulcer or glaucoma component; no ABDM/FHIR/ABHA integration (generic "EHR manager," not India's national framework); no federated learning; no real-time offline design.
DiabeticChain: A Novel Blockchain Approach for Patient-Centric Diabetic Data Management <i>The Journal of Supercomputing</i> , 2024 link.springer.com/article/10.1007/s11227-024-06589-6	<i>Proposes a permissioned, patient-centric blockchain architecture for diabetic data management, giving patients the power to grant or revoke access to their own diabetic data, addressing privacy, fragmentation, and interoperability problems in current centralized systems.</i>	Strong match on the "patient owns consent" model your ABDM/Consent Manager layer implements.	General diabetic data management, not eye-specific; no AI/CV screening component at all; not India/ABDM-specific.
Privacy-Preserving Technology Using Federated Learning and Blockchain in Protecting against Adversarial Attacks for Retinal Imaging <i>Ophthalmology</i> (AAO journal), 2025 aoajournal.org	<i>Develops federated deep learning algorithms across multiple hospital cohorts for detecting myopic macular degeneration from retinal photos and multi-disease classification from OCT, benchmarking against FedAvg, weighted FedAvg, and FedProx, and studying blockchain's role in defending FL against adversarial attacks.</i>	Directly validates your Layer 4 (federated learning) combined with blockchain, in ophthalmology specifically, from a high-impact clinical journal.	No edge deployment, no corneal component, no ABDM/consent-manager equivalent, no real-time single-patient screening workflow — this is a multi-institution research study, not a point-of-care product.
Federated Learning for Computer-Aided Diagnosis of Glaucoma Using Retinal Fundus Images <i>Applied Sciences</i> , 2023 doi.org/10.3390/app132111620	<i>Studies various federated learning algorithms and the impact of heterogeneous clients on federated glaucoma diagnosis performance, using public fundus image datasets from different sources to simulate different hospital clients.</i>	Validates federated learning specifically for glaucoma detection from fundus images — directly usable as your FL baseline methodology reference.	No blockchain component at all in this specific paper; no corneal/DR component; no real-world deployment/edge angle; authors explicitly note their API lacks production-grade security (SSL/encryption).
Deploying and Evaluating Multiple Deep Learning Models on Edge Devices for Diabetic Retinopathy Detection arXiv, 2025 arxiv.org/pdf/2506.14834	<i>Trains and quantizes MobileNet, ShuffleNet, SqueezeNet, and a custom DNN to TensorFlow Lite INT8 for DR detection on the EyePACS dataset, benchmarking inference speed, accuracy, and resource use across smartphones and microcontrollers; MobileNet reaches 96.45% accuracy, SqueezeNet reaches 17ms latency at 176KB.</i>	Directly validates your Layer 1 edge-AI approach and gives you concrete benchmark numbers to cite/compare against.	Pure edge-AI benchmarking paper — no blockchain, no EHR/ABDM integration, no corneal component, no consent management or FL.
Automated Detection of Corneal Ulcer Using Combination Image Processing and Deep Learning <i>Diagnostics</i> (MDPI), 2022 doi.org/10.3390/diagnostics12123204	<i>Proposes two corneal ulcer segmentation methods — classical image processing and deep semantic segmentation — using the public SUSTech-SYSU dataset of 712 fluorescein-stained slit-lamp images with 354 labelled for segmentation.</i>	Directly validates your Layer 1 corneal segmentation approach and is your primary source for the exact dataset and methodology to use.	No retinal/DR component, no blockchain, no EHR/ABDM integration, no real-time/edge deployment discussed.
Deep Learning Based Discrimination of Corneal Ulcer Patterns Using Fluorescein Staining Images ISICDM proceedings, ACM DL dl.acm.org/doi/10.1145/3451421.3451449	<i>Reports a segmentation dice similarity coefficient of 0.98 for cornea boundary and 0.97 for ulcer defect area, with 89.23% IoU for cornea segmentation and 82.94% IoU for ulcer segmentation on SUSTech-SYSU, plus a severity-grading accuracy of 86.15%.</i>	Gives you the concrete performance bar to beat/match for your U-Net corneal segmentation module.	Classification/grading focus, no blockchain, no ABDM, no retinal component, no real-time constraint discussed.
Real-world evaluation of smartphone-based AI to screen for diabetic retinopathy in Dominica <i>BMJ Open Ophthalmology</i> , 2023	<i>Prospective clinical validation of an AI system screening 587 diabetic patients in a real primary-care programme, reporting 77.5% sensitivity and 91.5% specificity for</i>	Gives you an honest, real-world clinical benchmark (not an inflated lab number)	No blockchain, no on-device/edge inference architecture detail, no consent/EHR integration, no

pmc.ncbi.nlm.nih.gov/articles/PMC10749075	referable DR versus a trained field grader (AUC 0.8455).	to cite for how well smartphone-based DR screening actually performs outside a lab.	corneal component.
Use of offline artificial intelligence in a smartphone-based [DR screening programme, Mumbai] <i>Indian Journal of Ophthalmology</i> , 2021 journals.lww.com/ijo	<i>Evaluated an offline AI algorithm on a smartphone across 1,378 diabetic patients in Mumbai public dispensaries, achieving 100% sensitivity and 89.55% specificity for referable DR versus grading by two vitreoretinal surgeons, emphasizing real-time offline analysis as useful in areas with limited internet.</i>	An India-specific, real-world validation of exactly the offline/edge philosophy your project uses — extremely relevant to cite as local-context justification.	No blockchain, no ABDM/ABHA (this predates ABDM's current maturity), no corneal component, no FL.
A Blockchain-Based Consent Mechanism for Access to Fitness Data in the Healthcare Context / Empowering Patients... Smart Contract-Enabled Informed Consent Strategy arXiv, 2022 & 2024 arxiv.org/pdf/2202.12582 , arxiv.org/pdf/2412.09820	<i>Surveys and proposes blockchain-based consent mechanisms — including MedRec, Ancile, ConsentChain, and Dovetail-style architectures — where smart contracts manage patient grant/revoke access to medical records, and proposes what its authors claim is the first blockchain/smart-contract system to manage informed consent specifically for clinical diagnosis and treatment decisions, not just data-sharing.</i>	Directly informs how to design your smart-contract-driven Consent Manager trigger for critical-case specialist referral (Section 5.2/5.3).	General consent-management research, not eye-disease specific, not India/ABDM-specific, no AI/CV component.

Bottom line on prior art: the Uppamma & Bhattacharya (2023) paper is close enough that you must cite and explicitly differentiate from it in your own report/paper — reviewers and your professor will likely know it, since it's a well-cited paper from an Indian institution (VIT) on almost the same idea. Your differentiators (edge deployment, corneal ulcer module, ABDM/FHIR-specific integration, federated learning, and consent-triggered urgent referral) are exactly what separates your work from theirs — make that comparison explicit and central in your own report's related-work section.

13. Scope, Novelty & Publishability Assessment

9.1 Is a paper possible from this, as-is?

Yes, realistically — but not by claiming "we combined blockchain and AI for eye screening," because as Section 12 shows, that specific claim has already been published (Uppamma & Bhattacharya, 2023, and several adjacent papers). A paper built on this topic is publishable at the level of a solid conference paper (IEEE conferences on healthcare informatics, or a decent Scopus-indexed journal) if you sharpen the claim to something none of the existing papers actually did. Below are concrete, honest novelty angles ranked by how strong/defensible they are.

Strong, defensible novelty angles

- The India-specific ABDM/FHIR integration itself.** None of the papers found combine an AI eye-screening pipeline with an actual working (even sandbox-level) ABDM HIP integration and FHIR `DiagnosticReport` generation. This is a genuinely under-explored, practically important gap — India-specific health-IT interoperability papers are valued precisely because they're grounded in a real national framework, not a generic "EHR."
- Multi-disease, multi-modality edge pipeline in one device.** Existing edge-AI DR papers do DR alone; existing corneal ulcer papers do corneal alone. Combining retinal classification and corneal segmentation in a single on-device pipeline, with a shared preprocessing stack, and reporting combined latency/accuracy/model-size numbers, is a legitimate systems-level contribution even if neither model architecture itself is new.
- Consent-triggered automated urgent referral.** The idea of a smart contract automatically initiating an ABDM Consent Manager request when a severity threshold is crossed — turning a screening result into an actionable, time-boxed specialist handoff — is not something the existing consent-management papers (Section 12) implement in an ophthalmology-specific, threshold-triggered way. This is a strong, narrow, defensible novelty claim.
- Empirical FL comparison specifically for the corneal ulcer task.** Federated learning has been studied for DR and glaucoma (Section 12), but not, as far as this search found, for corneal ulcer segmentation. Running FedAvg vs. FedProx on a partitioned SUSTech-SYSU-style setup for corneal segmentation would be a small but real, citable empirical contribution.

Recommended framing for your actual paper title/abstract: something like "An Edge-Deployed, ABDM-Compliant Blockchain Framework for Consent-Driven Ophthalmic Screening and Referral" — this signals the specific gap (ABDM compliance + consent-triggered referral automation) rather than the generic and already-published "blockchain + AI for eye disease" framing.

9.2 If the full system is too much to publish on, what's the minimum viable novelty?

If time runs out and you can't build all four layers fully, a paper is still very feasible from a subset:

- **Option A (safest):** Build only Layer 1 (edge AI, both corneal + retinal) really well, benchmark it thoroughly (accuracy, latency, model size across at least 2-3 device tiers), and frame the blockchain/ABDM/FL layers as "proposed architecture / future work" described in detail but not fully implemented. This is completely standard practice in student and even professional papers — a working core system plus a rigorously designed extension architecture.
- **Option B (more novel, more work):** Build Layers 1+2+3 (AI + blockchain + a *simulated* ABDM sandbox integration) fully, and describe Layer 4 (federated learning) as future work. This gets you the strongest unique claim (the ABDM integration) without needing multiple real hospital partners for FL.
- **Option C (if you want the FL angle specifically):** Drop the ABDM/blockchain layers to "described but not implemented," and instead spend your implementation time on a rigorous FL comparison (FedAvg vs FedProx vs a heterogeneity-aware variant) for corneal ulcer segmentation — genuinely under-studied per Section 12.

9.3 Where to submit

Realistic, achievable venues for a strong final-year project write-up: IEEE conferences on health informatics or biomedical engineering (many have student-friendly regional/national editions in India), Scopus-indexed journals like a suitable MDPI title (*Electronics, Diagnostics, Healthcare, Applied Sciences* — note these do carry article-processing charges, so check with your guide/department about funding), or your university's own student research symposium/journal as a first step before an external submission. Aim first for a solid conference paper; a journal submission with a novel angle can follow if the results are strong.

14. Suggested Implementation Roadmap (For a Final-Year Team)

Assuming a team of 3-4 students and roughly a 6-month major-project timeline, here is a realistic phase breakdown. Adjust based on your actual academic calendar.

Phase	Duration	Deliverables
1. Literature review + dataset acquisition	2-3 weeks	Finalize exact model architectures; download EyePACS/APTOS/IDRiD (retina) and SUSTech-SYSU (cornea); set up dev environment.
2. Retinal classification model	3-4 weeks	Train/fine-tune MobileNetV3 or EfficientNet-Lite for 5-class DR grading; evaluate accuracy, confusion matrix, sensitivity/specificity.
3. Corneal segmentation model	3-4 weeks	Train U-Net on SUSTech-SYSU; report Dice/IoU; compute ulcer-area percentage from mask.
4. Edge deployment	2 weeks	Convert both models to TFLite/ONNX, quantize, benchmark latency/size on at least one real Android phone.
5. Blockchain layer	2-3 weeks	Set up Hyperledger Fabric test network or Ganache+Solidity contracts; implement hash-logging and threshold-triggered events.
6. FHIR + ABDM sandbox integration	3 weeks	Set up HAPI FHIR server; generate valid <code>DiagnosticReport</code> resources; connect to ABDM's public developer sandbox for ABHA creation/linking (M1/M2 milestones).
7. Consent-triggered referral logic	1-2 weeks	Simulate the Consent Manager flow and specialist-access handoff (can be mocked if full HIE-CM sandbox access is unavailable).
8. Federated learning demo (optional/stretch)	2-3 weeks	Simulate 2-3 "hospital" nodes with partitioned data using Flower; compare FedAvg vs FedProx.
9. Dashboard + integration testing	2 weeks	React dashboard tying it all together; end-to-end demo run.
10. Report writing + paper drafting	2-3 weeks	Final report, optional paper draft, viva preparation.

Total: roughly 22-27 weeks, which fits a two-semester final-year project comfortably if Phase 8 is treated as a stretch goal rather than a hard requirement.

15. Risks, Limitations & Ethical/Regulatory Considerations

This is a screening aid, not a diagnostic replacement. Any report, demo, or paper arising from this project must clearly state

that the system flags patients for professional confirmation and does not replace an ophthalmologist's diagnosis. This is standard, expected language in every published paper reviewed in Section 12 and should be non-negotiable in your own framing.

- **Production ABDM access is gated.** Real production integration requires NHA empanelment, which is a lengthy compliance process — *certification typically takes 12 to 20 weeks* and is intended for actual healthcare providers, not student projects. Use the public developer sandbox and be explicit in your report that production deployment is future work.
- **Dataset bias and generalization.** Public datasets like EyePACS/APTOS are not necessarily representative of Indian population demographics or the specific camera hardware you'll use; report this limitation honestly rather than overstating accuracy.
- **Data protection law.** Any real patient data handling must be discussed against India's Digital Personal Data Protection (DPDP) Act, 2023 — for a student project, use only public, de-identified datasets; do not collect real patient images without institutional ethics clearance.
- **Blockchain overhead.** Writing to a blockchain has real latency and storage cost; be ready to defend this design choice against a simpler tamper-evident logging alternative (e.g., a signed, append-only database) in your viva — the honest answer is that blockchain adds value specifically when multiple mutually distrusting parties (hospital, patient, insurer, auditor) need to agree on a shared, unforgeable history, not simply for "innovation" points.
- **Federated learning realism.** Without real multiple hospital partners, your FL demo will necessarily be simulated (one machine, multiple partitions/processes). State this explicitly — it's completely acceptable practice, but don't imply real multi-hospital deployment happened.

16. Conclusion

This is a strong topic: clinically meaningful, technically substantial across multiple disciplines (computer vision, systems/blockchain engineering, health informatics), and grounded in a real, currently-expanding national digital health infrastructure. It is achievable as a major project if scoped honestly — build the AI and blockchain layers for real, treat the full-scale ABDM production integration and multi-hospital federated learning as simulated/future-work components, and be explicit and rigorous about that scoping in your report. The clearest path to a genuine publishable contribution is the ABDM/FHIR-specific integration and the consent-triggered urgent-referral mechanism, since neither appears in the closest existing published work. Cite the Uppamma & Bhattacharya (2023) paper directly and differentiate against it — that comparison alone will strengthen both your project report and any paper draft that follows from it.

17. References

Full source list (see Section 12 table for detailed context on each):

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15. ABDM FHIR Integration Guide 2026. Adrine. <https://www.adrine.in/blog/abdm-fhir-integration-guide-2026>

16. Federated Learning in Ocular Imaging: Current Progress and Future Direction. *Diagnostics*, 12(11), 2835 (2022).
<https://doi.org/10.3390/diagnostics12112835>

This report was compiled from publicly available research literature and technical documentation as of July 2026. Verify current ABDM API specifications and dataset licensing terms directly with official sources before implementation, as these evolve over time.